

SAHASRA JAVVADI

Full-Stack Engineer | Node.js · PostgreSQL · React

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EDUCATION

SRM University AP

BTech in Computer Science (CGPA: 9.28)

Andhra Pradesh, India

Oct 2022 – May 2026

WORK EXPERIENCE

Data Science Intern, Standard Chartered

July 2025 – May 2026

- Owned end-to-end backend engineering of a SharePoint automation service handling 100+ concurrent users — built validation, encoding edge-case handling, and concurrency controls in Python/Flask, then deployed to Dataiku production with zero post-launch manual intervention.
- Shipped a multilingual document ingestion pipeline processing 50+ PDFs across languages — engineered layout-aware table parsing with custom OCR post-processing to hit 92% accuracy, cutting manual processing time by 55% at scale.
- Designed and productionized a multi-dataset scoring pipeline joining audit plans, risk ratings, QA exceptions, and control logs — replaced a fully manual country-level audit process with programmatic risk-ranked surfacing across multiple geographies.
- Debugged production anomaly detection system on live audit datasets — traced and resolved systematic data quality failures that had evaded detection, reducing manual intervention by 40%.

PROJECTS

AI-Powered Interview Simulator | React, Node.js, Express.js, WebSocket, PostgreSQL

[GitHub](#)

- Architected and shipped a 4-module real-time full-stack app (React + Node.js/Express + PostgreSQL + WebSocket) — engineered 22+ REST API endpoints, resolved async race conditions breaking the live feedback loop, and handled concurrent WebSocket streams under load.
- Integrated GPT-4o-mini, Whisper, and TTS via OpenAI REST API — built rate-limiting, retry logic, and response-latency handling to deliver stable adaptive interview sessions at 3 difficulty levels without service degradation.

Student Counselling Management System | React.js, Node.js, Express.js, MongoDB

[GitHub](#)

- Designed 15+ RESTful APIs across a 3-role auth system (Admin/Student/Counselor) on Node.js/Express + MongoDB — engineered concurrent session handling for 100+ users and debugged auth state drift causing session corruption under load.
- Built full booking lifecycle across 4 status states with 20+ React components — resolved frontend-backend state sync failures under concurrent load and secured auth with Google reCAPTCHA integration.

AI Research Synthesizer | Python, LangChain, FAISS, FastAPI, Gemini API, RAG

[GitHub](#)

- Built an end-to-end RAG pipeline (LangChain + FAISS) ingesting research PDFs with chunked indexing and semantically accurate multi-document synthesis — gracefully handling retrieval failures and document parsing edge cases.
- Shipped a FastAPI backend with PDF ingestion, embedding-based top-k retrieval, and query-specific summarization — handling async I/O and query latency under iterative load testing.

PUBLICATIONS

Abstractive Text Summarization with Semantic and Contextual Alignment using BART [Paper](#) [Link](#)

2025 IEEE IConSCEPT | Fine-tuned BART on 300K+ articles achieving ROUGE-1: 58.33, ROUGE-2: 37.03, BERTScore F1: 86.31, outperforming 3+ baselines.

TECHNICAL SKILLS

- Backend & APIs:** Node.js, Express.js, Flask, FastAPI, REST API design & debugging, WebSocket
- Databases:** PostgreSQL, MySQL, MongoDB
- Languages:** Python, JavaScript, C++, SQL
- Frontend:** React.js, HTML/CSS
- Infrastructure:** Docker, Git, Dataiku, AWS (basics)
- CS Fundamentals:** Data Structures & Algorithms, OOP, Operating Systems, Computer Networks
- Familiar with:** NestJS (Express.js architecture; actively upskilling)

CERTIFICATIONS

- Building Language Models on AWS**
- Associate Developer – MongoDB**